

STATEMENT OF INSPECTION REVIEW AND APPROVAL

Part 1: TO BE COMPLETED BY THE DISTRICT OR OPERATING MSC UPON COMPLETION OF INSPECTION AND EVALUATION

A qualified inspection team, according to the NBIS and USACE Policy, has completed the inspection and evaluation of the CENNAPPA000004 / F.E. Walter Dam Service Bridge. The inspection report received an independent technical review by a qualified bridge engineer and all substantive issues arising during the review have been resolved to meet the provisions of PL 100-17 and existing USACE policy.

/s/ Joseph Gonglik
Bridge Inspection Team Leader

06-SEP-2022
(Final Report Signature Date)

28-JUL-2022
(Initial Report Signature Date)

/s/ Matthew Sosna
Technical Review Team Leader

07-SEP-2022
(Date)

/s/ Christine Clapp
Deputy Chief, E&C Division, NAP, NAD

07-SEP-2022
(Date)

Part 2: TO BE COMPLETED BY MSC

The inspection, evaluation and review of the F.E. Walter Dam Service Bridge has been completed in accordance with the approved Quality Control Plan, and this report is approved as sufficient in accordance with ER 1110-2-111 for updating the data in the CEBIS, and for proceeding to budget, schedule and design all maintenance and rehabilitation work necessary to correct deficiencies which jeopardize the structural safety and integrity of the F.E. Walter Dam Service Bridge.

/s/ Jiten Soneji
Chief, Engineering Division

08-SEP-2022
(Date)



U. S. Army Corps of Engineers

**F.E. Walter Dam Service Bridge
CENNAPPA000004**

**5 MI. EAST OF WHITE HAVEN
Luzerne County, Pennsylvania**

**NORTH ATLANTIC DIVISION
PHILADELPHIA DISTRICT**

INSPECTED APRIL 2022

1. Summary of Inspection:

Bridge Number: CENNAPPA000004

Bridge Location: Five miles east of White Haven, PA near PA Route 940 at the Pocono exit of the PA Turnpike Northeast Extension

Bridge Type: Steel two girder with concrete deck.

Inspection Date: 22 April 2022

Previous Inspection Date: 23 March 2017

Next Inspection: March 2027

Inspection Type: Routine/FCM Inspection

Overall Condition: Satisfactory-6

Deck-Previous Condition: Satisfactory-6

Deck-Current Condition: Satisfactory-6

Superstructure-Previous Condition: Good-7

Superstructure-Current Condition: Good-7

Substructure-Previous Condition: Satisfactory-6

Substructure-Current Condition: Satisfactory-6

Culvert-Previous Condition: N

Culvert-Current Condition: N

Channel-Previous Condition: Excellent-9

Channel-Current Condition: Excellent-9

Significant Recommendations: There were no recommendations that require immediate action; however, action should be prioritized based on the list of recommendations that can be found in the RECOMMENDATIONS section of the report.

New Critical Findings: None

Load Capacity: This bridge does not have a current load rating. A load rating will be completed in accordance with the district priority list and when funds become available.

Traffic Conditions: Access to this bridge is restricted. No traffic observed.

Inspection Team:

- Team Leader: Joseph Gonglik, P.E.–Philadelphia District
- Team Members: Robert Lawrence, P.E.–Philadelphia District

Note: The Team Leader meets the qualification requirements of the National Bridge Inspection Standards (NBIS) and ER 1110-2-111, USACE Bridge Safety Program.

Inspection Start-End Times: 0800–1100

Inspection Weather Conditions: Sunny and 65°F

Inspection Plan(s): See the Routine and Fracture Critical Plans located in the respective CEBIS INSPECTION REQUIREMENTS bridge files, for more detail on inspection procedures.

Procedures: The inspection did not deviate from the procedure described in the inspection plan.

- Access: Access for inspection of this bridge was by rope access techniques. A hands-on inspection was performed to review all accessible elements of the structure including fracture critical members. An underbridge inspection vehicle attempted to access the bridge but was too large. Only a smaller vehicle, such as an Aspen Aerial B-32 can access the bridge.
- Traffic Control: No traffic control measures were required for this inspection.
- Required Permits: No permits were required for this inspection.
- Inspection Methods: This inspection was conducted in accordance with the requirements of ER 1110-2-111. All exposed surfaces were visually inspected using the standard inspection tools following inspection procedures defined in the FHWA publication Bridge Inspector's Reference Manual (BIRM). Defect descriptions are based on the AASHTO Manual for Bridge Element Inspection, 1st Ed. An underwater inspection is not required.

Bridge Description: The bridge is approximately 420 feet long and consists of three riveted plate girder simple spans. Each span length is 138'-3", center-to-center of bearings (Drawing 1-1). The two girders are spaced 8'-0" apart center-to-center. Floorbeams are spaced approximately 11'-6" apart. The structure has top and bottom lateral bracing terminating at vertical cross bracing located at every other floorbeam. The reinforced concrete deck is 6-1/2" thick at centerline with a 1" crown and is composite with the steel girders. The roadway is 9 ft wide with curbs 9" high and 12" wide. Steel bridge railings are 3'-3" high, measured from the top of the curbs. Power and telephone conduits are attached to the inboard face of the girders. The center rail of both railings functions as an electrical conduit. The abutment and piers are constructed of reinforced concrete and are supported on spread footings. General views of the bridge are shown in Photos 1 and 2.

2. Summary of Findings:

Overall, the bridge is in good condition. No conditions were found during this inspection to indicate that the bridge is unsafe. Findings for specific components are as follows:

Deck: The bridge deck is in overall fair condition. There is scaling of the deck at both curb lines. Hairline transverse deck cracks are typical on the underside of the bridge deck and soffit with isolated areas of exposed rebar (see Photos 5 and 6). There are isolated areas of the bottom of the deck with delaminations, small spalls, honeycomb, and cracking (see Photo 7). The curb along Bay 4 of Span 3 exhibits exposed aggregate (see Photo 8).

Superstructure: The steel superstructure is in overall good condition. The superstructure exhibits only localized areas of crevice corrosion on secondary members (see Photo 21). Minor, isolated areas of pack rust developing on top flanges of exterior face of both girders (see Photos 9 and 10). Paint chipping with minor surface corrosion was noted on rivet heads and girder web in random locations. Some rivet and bolt heads exhibit pitting which was painted over in the previous painting contract. Rivets and bolts were randomly checked during this inspection and only one was found to be noticeably loose in Span 1 where a bolt on the lower lateral bracing gusset plate is visibly loose (see Photo 11).

Fracture Critical Members (FCM): The fracture critical members are in overall good condition (see Photo 20).

Substructure: The substructure is in overall satisfactory condition. The tops of the piers, corbel, and abutment were sealed to prevent water intrusion and further deterioration. Several spalled areas on piers were repaired in the previous painting and repair contract. Spalling, cracking, and efflorescence were noted at the pier caps and starting at the repaired areas (see Photo 12). A 10"x16"x2" spall has developed on the top of the pier cap of Pier 2 on the east face (see Photo 13). There is minor isolated efflorescence visible on the pier faces (see Photo 22).

Bearings: The bridge bearings are in overall fair condition. There is light, localized rust staining and lubrication is needed on all bearings. The rocker bearings at Pier 2 are leaning significantly in an over-expanded direction and the east and west girder bearings are shifted off of their keys (see Photos 16 and 17). There is significant section loss on some of the bearing anchor bolts and washers that was painted over in the previous paint contract (see Photos 18 and 19).

Approach Roadway: Very Good – 8, Equal to present desirable criteria. The asphalt approach roadway is in good condition.

Channel: Excellent - 9, There are no noticeable or noteworthy deficiencies which affect the condition of the channel.

Scour: 8, Bridge foundations determined to be stable for the assessed or calculated scour condition. Scour is determined to be above top of footing by assessment (i.e., bridge foundations are on rock formations that have been determined to resist scour within the service life of the bridge), by calculation or by installation of properly designed countermeasures (see HEC 23).

Waterway Adequacy: 9, Superior to present desirable criteria.

Safety Features: 0NNN

Complex Bridge Features: N/A

Underwater Inspection: N/A

Bridge Rail: Good – 7, The metal rail is in good condition.

3. Previous Inspection Findings and Recommendations:

The 2017 inspection report was available and used for comparison purposes.

Overall, the bridge condition is similar to the last inspection report. Since the last inspection, spalls have developed on the east faces of the pier caps of piers 1 and 2. The bridge has been cleaned and painted since the previous inspection. The previous report recommended:

- Replace pitted nuts and install spacers and washers to eliminate the gap between the anchor bolt nut and the shoe at the fixed bearings at Pier 2.
- Lubricate fixed bearing pins bi-annually.
- Lubricate rocker bearings bi-annually.
- Replace loose bolt at lower lateral bracing center gusset in Span 1, Bay 1.
- Clean and paint bridge with over coat system.
- Reset and modify the pier 2, east and west girder rocker bearing that is rocked off of the key.
- Perform Load Rating for bridge.

4. Recommendations:

The bridge is in good condition overall and experiences only low frequency use. Therefore, it is recommended that inspection of this bridge continue on a 5-year cycle. The following repairs are recommended to maintain the safety of the bridge and should be completed as soon as funding is available. The total estimated cost for all the repairs is \$63,700.

- Replace pitted nuts and install spacers and washers to eliminate the gap between the anchor bolt nut and the shoe at the fixed bearings at Pier 2. estimated cost: \$500.

Condition Classification	Component	Subcomponent
B	Bearings	Fixed Bearing

- Lubricate fixed bearing pins bi-annually. estimated cost: \$500.

Condition Classification	Component	Subcomponent
B	Bearings	Fixed Bearing

- Lubricate rocker bearings bi-annually. estimated cost: \$500.

Condition Classification	Component	Subcomponent
B	Bearings	Moveable Bearing (roller, sliding, etc.)

- Replace loose bolt at lower lateral bracing center gusset in Span 1, Bay 1. estimated cost: \$200.

Condition Classification	Component	Subcomponent
B	Secondary Members	Bracing

- Reset and modify the pier 2, east and west girder rocker bearings that have rocked off of the keys. estimated cost: \$50,000.

Condition Classification	Component	Subcomponent
B	Bearings	Moveable Bearing (roller, sliding, etc.)

- Perform Load Rating for bridge. estimated cost: \$7,000.

Condition Classification	Component	Subcomponent
A	General Comment	

- Patch spalled areas on east faces of pier caps at piers 1 and 2. estimated cost: \$5,000.

Condition Classification	Component	Subcomponent
B	Substructure	Reinforced Concrete-Pier Wall

5. Evaluation Summary:

The bridge is designed to support a H-15 truck. Continue monitoring conditions and conduct routine repairs as indicated within this report.

Load Rating: This bridge does not have a current load rating. A load rating will be completed in accordance with the district priority list and when funds become available.

Load Posting: Load Posting is not required.

Scour Evaluation: The bridge foundation is within the dam embankment, with no flow, therefore a scour evaluation is not required.

Seismic Evaluation: Seismic Category is B, which corresponds to structures in areas where expected ground shaking will be minor, and the soil is good. The Seismic Hazard Level (SHL) is sufficiently low where the bridge is likely not seismically vulnerable, and no further evaluation is necessary.

Fracture and Fatigue Evaluation: A fracture critical hands-on inspection of the steel girders was performed and the girders were found to be in overall good condition.

PUBLIC HIGHWAY/ROADWAY BRIDGES INVENTORY AND APPRAISAL (SI&A) SHEETS			Printed: 24-JUN-2025
202. Structure Number: CENNAPPA000004		8 NBI Structure Number: CENNAPPA-000004	
Geographical and Route Data		Dimensional Data	
1. State Code	Pennsylvania	32. Approach Roadway Width	9.0 feet
2. District	40	48. Length of Maximum Span	138.3 feet
3. County	Luzerne County	49. Structure Length	420.0 feet
4. Place Code	84552	53. Minimum Vertical Clearance Over Bridge Roadway	99.99 feet
6. Features Intersected	F.E. WALTER DAM	54. Minimum Vertical Underclearance	H 99.90 feet
7. Facility Carried by Structure	SERVICE BRIDGE	55. Minimum Lateral Underclearance on Right	H 99.7 feet
9. Location	5 MI. EAST OF WHITE HAVEN	56. Minimum Lateral Underclearance on Left	99.0 feet
16. Latitude	410642		
17. Longitude	0754315		
103. Temporary Structure Designation		Proposed Improvements	
On and Under Record Data		75. Type of Work	381
5. Inventory Route	180000000	76. Length of Structure Improvement	419.9 feet
10. Inv. Route, Mim. Vertical Clearance	99.99 feet	94. Bridge Improvement Cost	57
29. Average Daily Traffic	000001	95. Roadway Improvement Cost	0
30. Year of Average Daily Traffic	2022	96. Total Project Cost	64
47. Inventory Route, Total Horizontal Clearance	9.0 feet	97. Year of Improvement Cost Estimate	2022
General Data		114. Future Average Daily Traffic	1
12. Base Highway Network	0	115. Year of Future Average Daily Traffic	2038
22. Owner	70		
34. Skew	00	Appraisal Rating	
37. Historical Significance	5	67. Structural Evaluation	6
38. Navigation Control	0	68. Deck Geometry	4
42. Type of Service	56	69. Underclearances, Vertical and Horizontal	4
43. Structure Type Main	303	71. Waterway Adequacy	9
44. Structure Type Approach	000	72. Approach Roadway Alignment	8
45. No. of Spans Main	003	36. Traffic Safety Features	0NNN
46. No. of Approach Spans	0000	113. Scour Critical Bridges	8
27. Year Built	1961		
106. Year Reconstructed	0000	Load Rate and Post	
107. Deck Structure Type	1	41. Structure Open, Posted or Closed to Traffic	A
108. Wear/SurfProt Sys	100	63. Method Used to Determine Operating Rating	5
111. Nav Pier/Abut Protection		64. Operating Rating	19.40 ton
Condition Rating		65. Method used to Determine Inventory Rating	5
58. Deck	6	66. Inventory Rating	15.00 ton
59. Superstructure	7	70. Bridge Posting	5
60. Substructure	6		
61. Channel and Channel Protection	9	Over 200 Items	
62. Culverts	N	200. COE Division	CENAD
Inspection Data		201. COE District	CENAP
90. Inspection Date	04-2022	202. Number	CENNAPPA000004
91. Designated Inspection Frequency	60	203. Inspection Office	ECB
92. Critical Feature Inspection	Frac Crit Insp: Y60	204. Inspector	Joseph Gonglik, P.E.
	Underwater Insp: N	205. Inspections Cost	12000
	Other Spec Insp: N	216. Seismic Category	B
93. Critical Feature Inspection Date	Frac Crit Insp: 04-2022	217. Acceleration Coefficient	0.10
	Underwater Insp: -	218. Soil Site Coefficient	1.00
	Other Spec Insp: -	SR Sufficiency Rating:	49.2%

Structure Number
CENNAPPA000004

Name
F.E. Walter Dam Service Bridge

NBI Structure Number
CENNAPPA-000004

Operational Condition and Relative Risk Assessment

System:	Nonpublic Vehicle	231) Life Safety:	Low Impact (10)
235) Subsystem:	Outlet Works	232) USACE Mission:	Critical Impact (100)
233) Business Line:	Flood Risk Management	234) Regional Commerce:	Insignificant Impact (0)
236) Project (P2) number:	107566	Consequence Score/Category :	27 / II

Bridge Subcomponent Condition Classification

Component	Subcomponent	Condition Classification
Bearings	Fixed Bearing	B
	Moveable Bearing (roller, sliding, etc.)	B
Bridge Rail	Metal Bridge Railing	A
Deck	Reinforced Concrete Slab	B
Joint	Assembly Joint without Seal	A
Protective Systems	Steel Protective Coating	B
Safety Features	Lighting	A
Secondary Members	Bracing	B
Security Features	Fencing and Gates	A
	Lighting	A
Substructure	Reinforced Concrete-Abutment	B
	Reinforced Concrete-Pier Wall	B
Superstructure	Steel-Girder/Beam	B
Utilities/Attachments	Utilities	A

NBI No. CENNAPPA-000004

1. State: Pennsylvania
2. District: 40
9. Location: 5 MI. EAST OF WHITE HAVEN
6. Features Intersected: F.E. WALTER DAM

203. Inspection Office: ECB
204. Inspector: Joseph Gondlik, P.E.

Condition Ratings

58. Deck:	6
59. Superstructure:	7
60. Substructure:	6
61. Channel:	9

Appraisal Ratings

- 67. Structural Evaluation: 6
- 68. Deck Geometry: 4
- 71. Waterway Adequacy: 9
- 72. Approach roadway Alignment: 8
- 36. Traffic Safety Features: 0NNN
- 113. Scour Critical: 8

Element No.	Name	QTY	ENV	Cond 1	Cond 2	Cond 3	Cond 4
Decks/Slabs							
Span Range: 1 to 003							
12	Reinforced Concrete Deck	3761	1	0	3545	216	0
The bridge deck is in satisfactory condition. As previously reported, Transverse cracks are typical in the deck (Photo 3). Light to moderate scaling of the deck and parapet curbs exists at random locations. The condition of the leveling course in Span 3 has not changed since the previous inspection, exhibiting a smooth surface and transverse cracks at several locations (Photo 4). The roadway and drains are clean except for small accumulations of roadway gravel along the curbs and on the expansion joint at the abutment. It was previously reported that after rainfalls there are areas of standing water along both curbs. The caulking in the transverse contraction joints in the deck are in good condition.							
Transverse hairline cracks exist in the underside of the deck between adjacent floorbeams throughout the spans (Photo 5). Several of the cracks exhibit efflorescence. Moisture staining and honeycombing exist at random locations in the deck underside. Isolated areas of exposed rebar are also present (Photo 6).							
It is important to highlight that bay #4 from south to north of span 3 is more deteriorated than other bays along the bridge (Area = 108 SF). Not only the top of the deck on this bay #4, which was repaired for leveling course (Photo 4), but the underside of the deck shows delaminations, small spalls, honeycomb and cracking (Photo 7). Also the west curb of bay #4 shows severe exposed aggregate conditions (Photo 8).							
Small spalls and cracking are typical in random locations on the parapet overhangs. In the span 1 west overhang there is exposed rebar due to spalling with efflorescence and rust staining (Photo 9). There is also an area with pack rust and light spall at the east overhang of span 2 (Photo 10).							
Superstructure							
Span Range: 1 to 003							
107	Steel Open Girder/Beam	836	1	836	0	0	0
The steel girders are in good condition. The following deficiencies were noted:							
Span 1 Structural Steel							
Bay 1, lateral bracing lower gusset plate adjacent to tower corbel, loose nut in center (Photo 11).							
Span 2 Structural Steel							
- Bay 3, at the southeast corner, two loose rivets at the fill plate in the cross bracing. Previously reported. - Bay 4, 1" gouge found in the interior face of the east girder, underneath the gusset angle. No distress was noted. - Bay 6, a visible gouge in the interior face of the east girder. The gouge is located approximately 2 feet below the top flange of the girder, and is approximately 1-1/2" in length. No distress was noted.							
Span 3 Structural Steel							
- Bay 5, the web and southeast and southwest stiffeners are out-of-plane. There were no signs of distress, but this area should be monitored in future inspections. - Bay 6, section loss in the floorbeam bottom flange measuring 1-1/4" by 3/8". The section loss is approximately 3/8" deep. No immediate signs of distress were apparent in the floorbeam during the inspection.							
Span Range: 1 to 003							
107	Steel Open Girder/Beam Paint System	836	1	761	75	0	0
The condition of the paint cover is good (Photo 9). There is minor rust staining from crevice corrosion on some of the lateral bracing connections (Photo 21).							
Substructure							
Span Range: 1 to 003							
210	Reinforced Concrete Pier Wall	30	1	24	6	0	0
Pier 1 is in satisfactory condition.							
- On the south face, approximately 18" below the second construction joint, a minor crack of about 1/16" thickness, and 6 ft long was noted. - On the south face, between construction joints 4 and 5, towards the east, delamination of patchwork was noted. - Also on the south face, a 6" by 1" spall exists below the 6th construction joint. - Shallow scaling exists on the other three sides of the pier between the second and third construction joints. - Pier Cap 1, on the South face, a 2' by 3' delamination of patchwork and efflorescence was noted in the previous inspection. - The east face of pier cap 1 is exhibiting spalling and cracking with efflorescence (Photo 12).							
Pier 2 is in satisfactory condition.							
- On the south face of the pier cap, efflorescence was noted below both bearings. - On the north face of the pier cap, efflorescence was noted below the bearing, on the exterior and interior sides of bearing along the height of pier cap. - On the east face a 10"x16"x2" spall was noticed at top of pier cap (Photo 13). - On the North face, above construction joint 5, three rebar ends are exposed.							
Span Range: 1 to 003							
210	Reinforced Concrete Pier Wall Tower Corbel	30	1	30	0	0	0
The bearing area for Span 1 was examined and is in good condition.							
Span Range: 1 to 003							
215	Reinforced Concrete Abutment	30	1	30	0	0	0
The abutment is in good condition. The bituminous sealer in the contraction joints continues to be cracked and separated (Photo 14). The surfaces of the abutment and wingwalls are scaled. Other unsound areas were found on the south half of the west wingwall at the abutment. A small edge spall at the top of the abutment, west cheekwall was repaired (Photo 15).							
Bearings							
Span Range: 1 to 003							
311	Movable Bearing	6	1	4	0	2	0
The expansion rocker bearings are in overall fair condition.							
The southwest anchor bolt in the east rocker bearing at Pier 1 is not vertically aligned.							
The rocker bearings at Pier 2 are in good condition; however, the rocker bearing of the west girder continue to lean significantly in the expanded position at relatively low temperatures as reported in previous inspection. The rocker is shifted on the base plates and is bearing partially on the end retainer teeth (Photo 16). On the other side, the east							

girder rocker bearing is completely rocked off of the key (Photo 17). The top surfaces of the teeth are worn, apparently from contact with the rockers. The bearings are stable in there current condition. The overexpanded bearings should be reset and modified to allow for normal thermal expansion and contraction. There has been no change since the previous inspection.

The rocker bearings at the abutment are in good condition.

All rocker and fixed bearing pins need lubrication.

Rocker bearing measurements were measured during previous inspections and are tabulated on Drawing 1-2. The measurements indicate normal movement, allowing for the over-expanded rockers at Pier 2. It is recommended that future measurements be made at a temperature consistent with either one of the two previous measurements. It is also recommended that the rocker bearings be lubricated at both piers.

Span Range: 1 to 003

313 Fixed Bearing	6	1	6	0	0	0
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The fixed bearings are in overall good condition. The west anchor bolts in the east fixed bearing at Pier 1 have no washers beneath the nuts covering the holes through the masonry plate (Photos 18 and 19). The anchor bolts in the fixed bearings at Pier 2 are not vertical (Photo 16). The anchor bolt nuts are not tightened down and there is a gap between washers and the bottom of the shoe. It is recommended that spacers and washers be installed in the gap.

Bridge Railings

Span Range: 1 to 003

330 Metal Bridge Railing	836	1	836	0	0	0
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The bridge railing is in good condition. The railing and other miscellaneous steel above the deck was painted since 2017. Minor isolated rust staining is now present (Photo 23).

Joints

Span Range: 1 to 003

303 Assembly Joint With Seal	36	1	36	0	0	0
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The roadway joints at the piers and tower are in good condition. The abutment joint includes dirt and gravel in the sliding plate joint crevice. Roadway joint measurements taken during previous inspections are tabulated on Drawing 1-3.



Photo 1: East elevation of bridge.



Photo 2: General view of deck looking towards tower.



Photo 3: Typical transverse crack on deck.

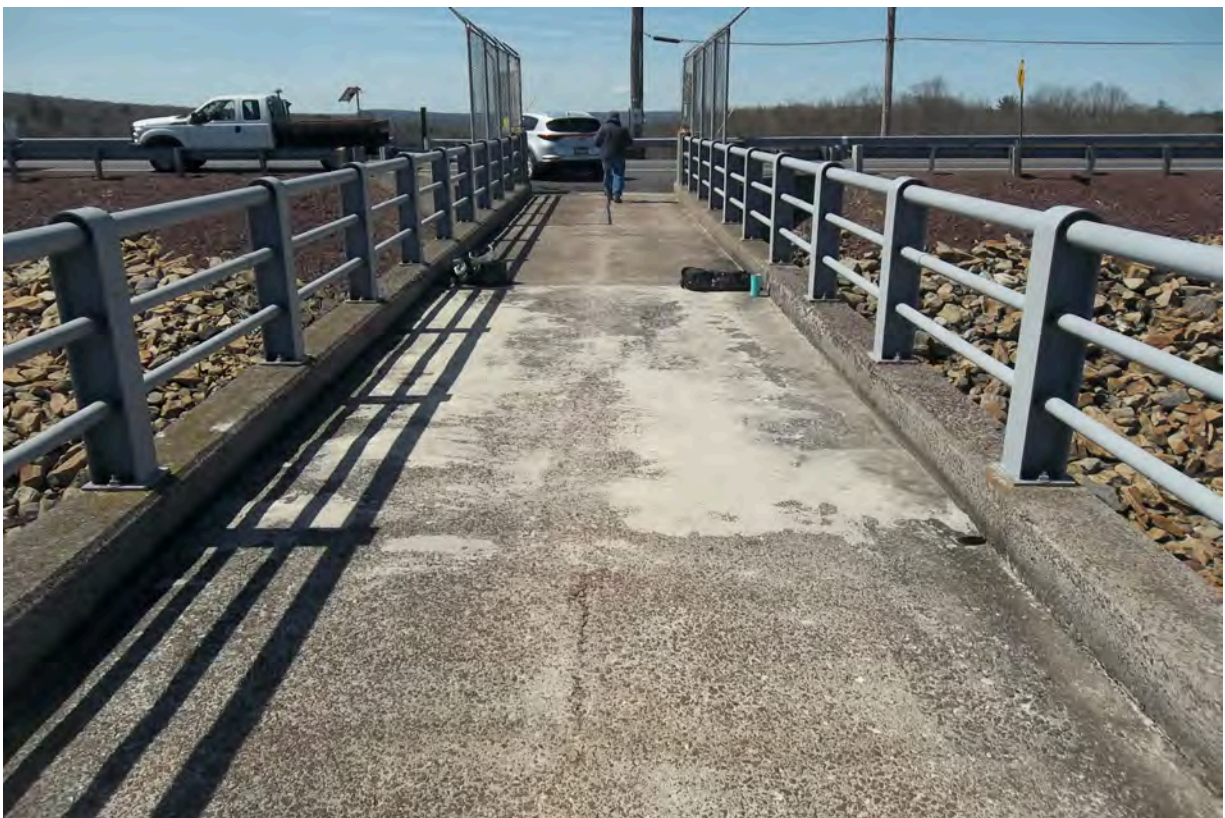


Photo 4: View of concrete deck repair for leveling course on span 3.



Photo 5: Transverse cracking in deck soffit.



Photo 6: Cracks in underside of deck with isolated exposed rebar.

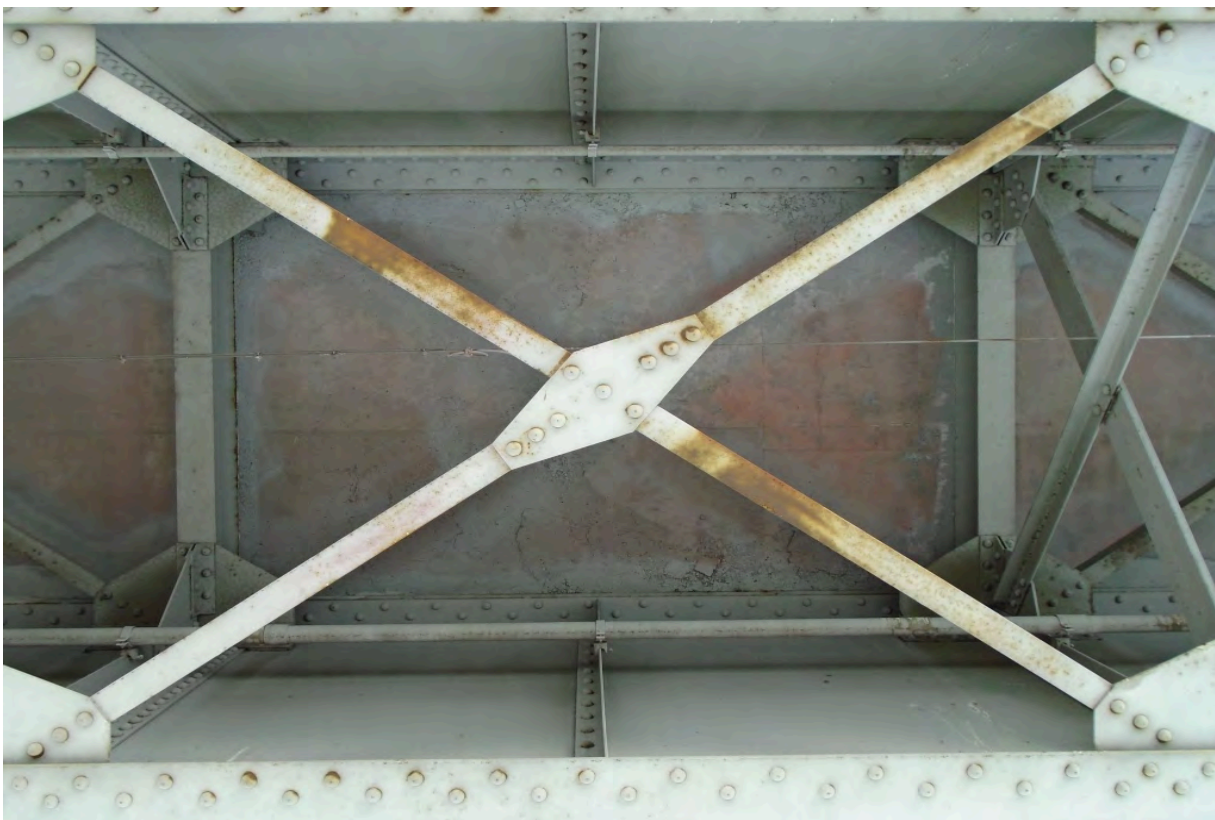


Photo 7: Bottom of deck with delaminations, small spalls, honeycomb, and cracking at bay #4 from south to north of span 3 (photo from 2017).



Photo 8: Typical exposed aggregate condition of curb along bay #4 from south to north of span 3 (photo from 2017).



Photo 9: Pack rust and spalling with efflorescence and rust staining at west overhang of span 1.



Photo 10: Pack rust and light spalling at the east overhang of span 2.



Photo 11: Span 1, bay 1, lateral bracing lower gusset plate adjacent to corbel, loose nut in center.



Photo 12: Pier 1, east face spalling and cracking with efflorescence on pier cap below east bearing.



Photo 13: Pier 2, 10"x16"x2" spall at top of pier cap, east face.



Photo 14: South abutment, 1/8in-1/4in horizontal crack on the east wingwall. Missing vertical joint material 1/8in (photo from 2017).



Photo 15: South abutment, repaired 10in long spall on the west cheekwall. Hairline to 1/16in horizontal crack on the west wingwall (photo from 2017).



Photo 16: Pier 2, west fixed bearing from span 3, no lubrication, anchor bolts are leaning south. Note rocker bearing has rocked off key.



Photo 17: East view of east rocker bearing from span 2 and pier 2. Rocker bearing has totally rocked off of key.



Photo 18: Span 2, pier 1, east fixed bearing, west anchor bolts are missing washers. Note section loss on east nuts.



Photo 19: Pier 1, fixed bearing anchor bolts have loose nuts and no washers.



Photo 20: Typical condition under bridge between girders.



Photo 21: Typical crevice corrosion on interior of bridge girders and floorbeams.



Photo 22: View of north face of pier 1. Minor isolated efflorescence visible.



Photo 23: Typical rail condition, paint with minor isolated rust staining.

Additional Information may be found at:

[DRAWING 1-0 VICINITY MAP](#)

[DRAWING 1-1 PLAN AND ELEVATION](#)

[DRAWING 1-2 EXPANSION BEARING MEASUREMENTS](#)

[DRAWING 1-3 ROADWAY JOINT MEASUREMENTS](#)

[F.E. WALTER SERVICE R&A SUMMARY 2022](#)